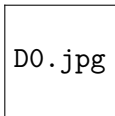


Lecture 45: Cloud Databases — AWS, Azure, Google Cloud, and Supabase

Dr. Innocent Nyalala



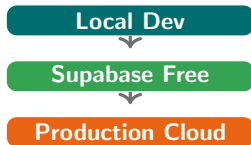
IIT Madras Zanzibar

22 June 2026

Lecture Outline

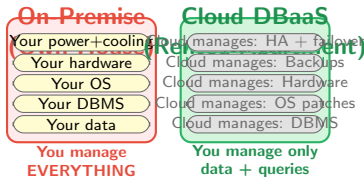
- 1 Why cloud databases?
- 2 Major cloud DB services
- 3 Supabase: your project platform
- 4 Managed vs. serverless
- 5 Cost models and free tiers
- 6 Multi-region replication
- 7 Project deployment guide
- 8 Course wrap-up

From your laptop to the cloud:



Z2004 Project pathway

Why Cloud Databases? The Apartment Analogy



Cloud DB Advantages

- ▶ No hardware procurement
- ▶ Automated backups built-in
- ▶ Scale up/down in minutes
- ▶ High availability by default
- ▶ Pay-as-you-go pricing
- ▶ Global edge locations

On-Premise Still Needed When

Regulatory compliance (data cannot leave country), extreme low-latency (<1ms), sensitive government/military data, or highly customized hardware requirements.

Major Cloud Database Services: The Landscape

Cloud	Relational	NoSQL	Specialty
AWS	RDS, Aurora	DynamoDB	Redshift (DW)
Azure	Azure SQL	Cosmos DB	Synapse (DW)
Google	Cloud SQL	Firestore	BigQuery (DW)
Supabase	PostgreSQL	—	pgvector, AI
PlanetScale	MySQL-compatible	—	Serverless scale
Neon	PostgreSQL	—	Serverless, branches

AWS
Amazon RDS, Aurora, Amazon Redshift

Azure
Azure SQL Database, Cosmos DB (multi-model)

Google
Google Cloud SQL, Firestore, BigQuery

For Z2004 Project

Recommended path:

- 1 Local PostgreSQL (development)
- 2 Supabase free tier (demo + grading)
- 3 AWS RDS (if scaling beyond free)

Free Tiers

All major clouds offer generous free tiers for students. Supabase free: 500MB storage, 2GB bandwidth, 50K API requests/day

Supabase: Firebase Alternative on PostgreSQL

Supabase = PostgreSQL + Superpowers

PostgreSQL Full Full SQL, transactions, ACID

pgvector Semantic Semantic search for your RAG project

Auto REST API REST API from your schema (no code!)

Auth & RBAC Row-level security, user management

Storage S3-compatible file storage (PDFs, images)

Realtime Socket subscriptions on table changes

Get this URL from Supabase dashboard

```
DATABASE_URL = "postgresql://postgres:[PASSWORD]@db.[project].supabase.co:5432/postgres"
conn = psycopg2.connect(DATABASE_URL) cur = conn.cursor()
Works exactly like local PostgreSQL! cur.execute("SELECT
COUNT(*) FROM chunks;") print(cur.fetchone())
Enable pgvector cur.execute("CREATE EXTENSION IF NOT EXISTS
vector;") conn.commit()
```

Setup in 5 minutes

supabase.com → New Project → Copy
connection string → replace
LOCAL_DB_URL in your code. Done!

Managed vs. Serverless Databases

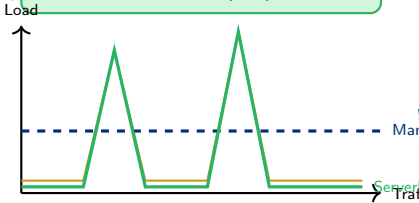
Always-On vs. Scale-to-Zero

Managed (Always On)

Instance always running. Pay per hour. Predictable cost. Fast cold start (0ms). Examples: PostgreSQL, MySQL, Oracle.

Serverless (Scale to Zero)

Scales to zero when idle. Pay per request. Slow cold start (1–5s). Examples: Aurora Serverless, Supabase, Neon.



Property	Managed	Serverless
Cold start	0ms	1–5s
Idle cost	Always pay	Free
Predictable cost	Yes	Spiky
Max throughput	Limited	Auto-scaled
Connections	Always ready	May need to wait
Best for	Production	Dev/staging

Project Advice

Use Supabase free tier (managed): always-on, 0ms cold start, easy pgvector. For demo: reliable performance matters more than cost.

Cost Models: What Does It Actually Cost?

Student Project Cost Comparison (per month)

Supabase Free	1GB, 2GB bandwidth, pgvector	\$0
Neon Free	512MB, 190 compute hours	\$0
AWS RDS Free	750hrs, 20GB, 12 months	\$0*
Supabase Pro	8GB, unlimited bandwidth	\$25
AWS RDS t3.micro	20GB, always-on	\$15
AWS RDS t3.small	20GB, always-on	\$28

* AWS free tier: first 12 months only

Free Tier Strategy

- ▶ Supabase: best for pgvector + easy setup
- ▶ Neon: best if you need branching (like git for databases!)
- ▶ AWS: best if you're learning AWS ecosystem
- ▶ All have generous free tiers for students

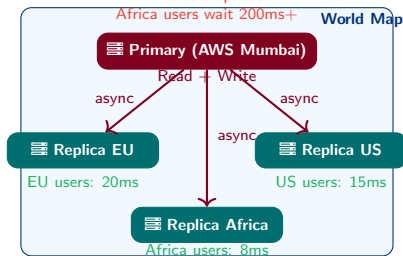
Cost Optimization Tips

- ▶ Connect pooling (pgbouncer) saves connections

▶ Store indexes efficiently (HNSW)

Multi-Region Replication: Global Performance

Primary + Read Replicas Architecture



Replication Benefits

- ▶ Users read from nearest replica
- ▶ Reduced read latency (global)
- ▶ Writes go to primary only
- ▶ Async replication: slight lag
- ▶ Sync replication: consistent but slower

Supabase Read Replicas

Supabase Pro supports read replicas in multiple regions. For \$25/month, your IITMZ project could serve students in Zanzibar, India, and globally with low latency.

Connecting Your Project to Supabase

```
import psycopg2 import os from sentence_transformers import SentenceTransformer
Step 2: Connect (use env variables, not hardcoded!)
DATABASE_URL = os.environ.get(
'SUPABASE_DB_URL') conn = psycopg2.connect(DATABASE_URL) cur = conn.cursor()
Step 3: Enable pgvector (run once) cur.execute("CREATE EXTENSION IF NOT
EXISTS vector;") conn.commit()
Step 4: Create your table cur.execute(""" CREATE TABLE IF NOT EXISTS
Chunks ( id SERIAL PRIMARY KEY, source VARCHAR(200), content TEXT,
embedding vector(384) ); CREATE INDEX IF NOT EXISTS
chunks_embedding ON Chunks USING hnsw (embedding vector_cosine_ops); """) conn.
print("Supabase connected and ready!")
```

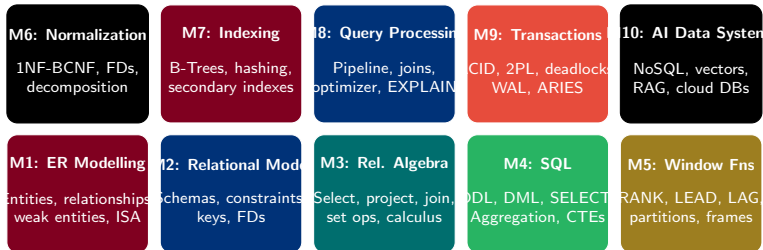
Deployment Steps

- 1 Create free Supabase account
- 2 New project (choose Singapore region, closest to Zanzibar)
- 3 Copy connection string
- 4 Set as env variable
SUPABASE_DB_URL
- 5 Run schema setup script
- 6 Run ingestion pipeline
- 7 Test with 5 queries
- 8 Share URL for grading

Security Rule

Skills You've Gained: The Full Picture

Z2004 Database Management Systems — Complete Skills Map



Every module connects. Real systems use ALL of these.

What's Next Beyond Z2004?

Graduate-Level Opportunities at IITMZ

Z5008 Big Data Lab (next semester!)

Cassandra, Spark, Hadoop, streaming

Z5007 — ML and Deep Learning

ML on top of your DBMS skills

Distributed Systems CS6xxx (future)

Rate, Paxos, distributed transactions

SAAIL Lab Research Open positions

Semantic search, agricultural AI, RAG

Industry — FAANG/startups

Data engineer, ML engineer, DB admin

Z5008 Big Data Lab

Starting March 2026 (MTech). Topics:

- ▶ Apache Cassandra at scale
- ▶ Apache Spark (distributed SQL)
- ▶ Kafka (streaming data)
- ▶ Hadoop ecosystem
- ▶ Real industry datasets

Z2004 is your foundation for this!

SAAIL Lab Openings

Looking for MTech and Research Assistants for projects in: AI-powered agricultural data systems, satellite image

Good Luck — You've Got This!

End-Semester Exam Preparation

ER Modelling + Relational Schema 15%

SQL Queries (all forms) 20%

Normalization (FDs, 2NF, 3NF, BCNF) 15%

Indexing (B-Tree, Hash, EXPLAIN) 15%

Query Processing + Join Algorithms 15%

Transactions, ACID, 2PL, Deadlocks 10%

NoSQL + Vectors + Cloud (L42–L45) 10%

Final Reminders

- ▶ Project Milestone 1 due soon
- ▶ Office Hours: **Wednesday 2–4pm, Room 105**
- ▶ Email: innocent@iitmz.ac.in
- ▶ No extension without medical certificate

“The best database is the one you actually understand.”

You now understand them deeply.

Good luck! — Dr. Nyalala



Class of Z2004, March–July 2026